CO2 -ASSESSMENT TOOL 3

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

SCENARIO BASED QUESTIONS

Q1. Data Preprocessing in Retail Analytics (*L3, L4)*

A **retail analytics company** is preparing customer purchase data for building a recommendation system. The dataset contains missing values, inconsistent categorical formats, and extreme values due to data collected from multiple stores and manual entries.

Sample Dataset

Customer_ID	Age	Gender	Monthly _Spend (₹K)	Purchase _Frequency	Loyalty _Level
C1	23	M	20	5	Low
C2	NaN	Male	35	8	Medium
C3	31	F	NaN	10	High
C4	105	Female	500	50	High
C5	27	male	25	NaN	Low
C6	40	FEMALE	45	12	Medium
C7	36	M	38	9	Low
C8	42	Female	300	45	High

Tasks

A. Handling Missing Values
i. Identify missing values in **Age, Monthly_Spend, and Purchase_Frequency**
ii. Apply at least **two imputation techniques** (e.g., Median Imputation, k-NN Imputation)
iii. Compute the imputed values and justify the selected method

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B. Outlier Detection and Treatment
i. Detect outliers in **Age and Monthly_Spend** using the **Z-score method**
ii. Identify records containing extreme values (e.g., Age = 105, Spend = 500)
iii. Apply suitable treatment methods (**capping / transformation / removal**) and justify

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C. Categorical Data Standardization
i. Identify inconsistencies in **Gender** (M, Male, male, FEMALE, etc.)
ii. Convert all entries into a standardized format (**Male / Female**)

Q2. Covariance & Data Reduction in E-commerce Analytics *(L4, L5)*
An **e-commerce platform** is building a predictive model to classify high-value customers. The dataset includes several correlated financial and behavioral attributes, leading to redundancy and increased computational cost.
Sample Dataset

Cust_ID	**Age**	**Income (₹K)**	**Purchase_ Value (₹K)**	**Credit_ Score**	**Orders**	**Avg_Balance (₹K)**	**EMI (₹)**	**Customer_ Class**
U1	24	40	100	650	15	12	5000	Low
U2	34	60	150	700	20	18	7000	Medium
U3	44	85	210	720	28	25	9000	Medium
U4	52	120	320	800	35	40	15000	High
U5	29	45	110	680	18	14	5500	Low
U6	39	75	180	710	24	22	8500	Medium
U7	56	150	360	820	40	50	18000	High
U8	37	70	160	705	22	20	8000	Medium

Tasks
A. Covariance Analysis
i. Compute covariance between **Income and Purchase_Value** (show steps or partial calculation)
ii. Interpret whether the relationship is **positive or negative**
iii. Explain how covariance helps in identifying feature relationships

B. Feature Selection
i. Identify redundant attributes using **correlation or domain knowledge**
ii. Select an optimal subset of features for modeling
iii. Justify your selection

C. Data Transformation
i. Standardize the dataset (excluding **Cust_ID and Customer_Class**)
ii. Explain why normalization/standardization is required

Q3. Data Transformation & Discretization in Education Analytics *(L3, L4, L5)*

An **education analytics team** is developing a model to predict student performance. Continuous variables such as study hours and test scores need to be discretized to improve interpretability and decision-making.

Sample Dataset

Student_ID	Study_Hours	Test_Score	Attendance (%)	Performance
S1	2	40	60	Poor
S2	3	50	65	Average
S3	4	55	70	Average
S4	5	65	75	Good
S5	6	75	80	Good
S6	7	85	85	Good
S7	8	90	88	Excellent
S8	9	95	90	Excellent
S9	10	98	92	Excellent
S10	11	100	95	Excellent

Tasks

A. Equal-Width Binning
i. Apply **3 bins** for Study_Hours and Test_Score
ii. Calculate bin width and define intervals

B. Equal-Frequency Binning
i. Divide data into **3 bins with equal records**
ii. Assign values accordingly

C. Concept Hierarchy Construction

- Study_Hours → Low / Medium / High
- Test_Score → Poor / Average / Excellent

Q 4. Use the two methods below to normalize the following group of data:
200, 300,400, 600, 1000
(a) min-max normalization by setting min = 0 and max = 1
(b) z-score normalization

Q 5. Data Transformation in Online Shopping Behavior Analysis *(L3, L4, L5)*
An **e-commerce company** is analyzing customer engagement by measuring the **time spent (in seconds)** on product pages. The duration varies significantly—from a few seconds to more than an hour—resulting in highly skewed data.
To improve the effectiveness of **customer segmentation using clustering techniques**, the dataset must be properly transformed and normalized.

Sample Dataset: Visit Duration

Customer_ID	Visit_Duration (seconds)
C1	15
C2	30
C3	45
C4	60
C5	120
C6	300
C7	600
C8	900
C9	1800
C10	3600

Tasks

A. Equal-Frequency Binning

i. Divide the dataset into **3 bins** with approximately equal number of records

ii. Assign each value to its respective bin

iii. Analyse how effectively this method groups short and long visit durations

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B. Z-Score Normalization

i. Compute the **mean and standard deviation** of visit duration

ii. Apply Z-score normalization:

$$Z = \frac{X - \mu}{\sigma}$$

iii. Discuss how this method handles **extreme values (e.g., 1800, 3600 seconds)**

C. Min-Max Normalization

i. Apply Min-Max normalization using range **[0,1]**:

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}}$$

ii. Transform all values accordingly

iii. Compare results with Z-score normalization in terms of **scaling and sensitivity to outliers**

Answers:

1) A retail analytics company is preparing customer purchase data for building a recommendation system.
The dataset contains:

- Missing values
- Inconsistent categorical formats
- Extreme (outlier) values

A. Handling Missing Values
i. Identify Missing Values

Attribute	Missing Record
Age	C2
Monthly_Spend	C3
Purchase_Frequency	C5

ii. Apply Two Imputation Techniques
Method 1: Median Imputation
Age Values
23, 31, 105, 27, 40, 36, 42
Sorted:
23, 27, 31, **36**, 40, 42, 105
Median = **36**
Therefore
Missing Age (C2) = 36

Monthly Spend
20,35,500,25,45,38,300
Sorted
20,25,35,**38**,45,300,500
Median = **38**
Therefore
Missing Monthly Spend (C3)= **38**

Purchase Frequency
5,8,10,50,12,9,45
Sorted
5,8,9,10,12,45,50
Median = **10**
Therefore
Missing Purchase Frequency (C5)= **10**

Dataset after Median Imputation

Customer	Age	Monthly Spend	Purchase Frequency
C1	23	20	5
C2	36	35	8
C3	31	38	10
C4	105	500	50
C5	27	25	10
C6	40	45	12
C7	36	38	9
C8	42	300	45

Method 2: k-NN Imputation
The k-Nearest Neighbor algorithm estimates missing values using the nearest records based on similarity.
Example:
For Customer C2
Nearest customers:

- C6 (Age = 40)
- C7 (Age = 36)
- C3 (Age = 31)

Average
=(40+36+31)/3
=35.67 ≈ **36**
Similarly,
Monthly Spend ≈ 36–38
Purchase Frequency ≈10

iii. Justification
Median Imputation
Advantages

- Works well with outliers
- Simple
- Preserves middle value

k-NN Imputation
Advantages

- Uses similar customers
- More accurate
- Preserves relationships among attributes

B. Outlier Detection using Z-Score
Formula

$$Z = \frac{x - \mu}{\sigma}$$

where

- μ = Mean
- σ = Standard deviation

Step 1: Age
Values
23,36,31,105,27,40,36,42
Mean

$$\mu = \frac{340}{8} = 42.5$$

Standard deviation
≈24.8

Z-Scores

Age	Z-score
23	-0.79
36	-0.26
31	-0.46
105	2.52
27	-0.63
40	-0.10
36	-0.26

Age	Z-score
42	-0.02

Since
Age =105
has the highest Z-score, it is considered an **outlier**.

Step 2: Monthly Spend
Values
20,35,38,500,25,45,38,300
Mean
125.13
Standard deviation
≈173.6

Z-Scores

Spend	Z-score
20	-0.61
35	-0.52
38	-0.50
500	2.16
25	-0.58
45	-0.46
38	-0.50
300	1.01

The highest value
₹500K
is treated as an outlier.

ii. Records Containing Outliers

Customer	Outlier
C4	Age=105
C4	Monthly Spend=500

iii. Treatment Methods
Method 1
Cap Age
105 → 60

Method 2
Cap Monthly Spend
500 →300

Method 3
Remove Record
Delete Customer C4 if it is caused by data entry error.

Justification
Capping is preferred because it preserves the record while reducing the impact of extreme values.

C. Categorical Data Standardization
Original values

Original	**Standardized**
M	Male
Male	Male
male	Male
Female	Female
FEMALE	Female
F	Female

Standardized Dataset

Customer	**Gender**
C1	Male
C2	Male
C3	Female
C4	Female
C5	Male
C6	Female
C7	Male
C8	Female

The dataset has been successfully preprocessed by:
- Handling missing values using Median and k-NN imputation.
- Detecting outliers using the Z-score method.
- Treating extreme values through capping or removal.
- Standardizing categorical values into consistent formats.

This preprocessing improves the quality and reliability of the data for recommendation system development.

2) **A. Covariance Analysis**
Formula

$$Cov(X, Y) = \frac{\sum (X - \bar{X})(Y - \bar{Y})}{n - 1}$$

Income
40,60,85,120,45,75,150,70
Purchase Value
100,150,210,320,110,180,360,160

Step 1
Mean Income

$$\bar{X} = \frac{645}{8} = 80.625$$

Mean Purchase Value

$$\bar{Y} = \frac{1590}{8} = 198.75$$

Step 2

Income	**Purchase**	**(X−X̄)**	**(Y−Ȳ)**	**Product**
40	100	-40.63	-98.75	4011.72
60	150	-20.63	-48.75	1005.47
85	210	4.38	11.25	49.22
120	320	39.38	121.25	4774.22
45	110	-35.63	-88.75	3161.72
75	180	-5.63	-18.75	105.47
150	360	69.38	161.25	11185.16
70	160	-10.63	-38.75	411.72

Total Product
≈24699

Covariance

$$Cov = \frac{24699}{7}$$

= **3528.43**

ii. Interpretation
Covariance is **positive**.
Therefore,
- Higher income leads to higher purchase value.
- Both variables increase together.

iii. Importance of Covariance
Covariance helps to
- Identify relationships among features
- Detect redundant variables
- Improve feature selection
- Reduce dimensionality

B. Feature Selection
Highly Related Features
- Income
- Purchase Value
- Average Balance
- Credit Score

These variables contain similar information.

Selected Features
- Age
- Income
- Orders
- Credit Score
- Customer Class

Justification
These features contain sufficient information while reducing redundancy and computational complexity.

C. Data Standardization
Formula

$$Z = \frac{x - \mu}{\sigma}$$

Example for Income
Mean
80.63
Standard deviation
36.83
For Income=40

$$Z = \frac{40 - 80.63}{36.83} = -1.10$$

Similarly

Income	Standardized Value
40	-1.10
60	-0.56
85	0.12
120	1.07
45	-0.97
75	-0.15
150	1.88
70	-0.29

The same process is repeated for all numerical attributes except:

- Cust_ID
- Customer_Class

Why Standardization?

- Removes scale differences.
- Improves machine learning performance.
- Speeds up model training.
- Gives equal importance to all attributes.
- Prevents large-valued features from dominating the model.

3) **Given Dataset**

Student	Study Hours	Test Score	Attendance	Performance
S1	2	40	60	Poor
S2	3	50	65	Average
S3	4	55	70	Average
S4	5	65	75	Good
S5	6	75	80	Good
S6	7	85	85	Good
S7	8	90	88	Excellent
S8	9	95	90	Excellent
S9	10	98	92	Excellent
S10	11	100	95	Excellent

A. Equal-Width Binning
(i) Study Hours
Minimum = 2
Maximum = 11
Range
=11−2

=9
Number of bins = 3
Bin Width

$$\frac{11-2}{3} = 3$$

Intervals

Bin	**Interval**
Bin 1	2–5
Bin 2	5–8
Bin 3	8–11

Assignment

Study Hours	**Bin**
2	Bin 1
3	Bin 1
4	Bin 1
5	Bin 1
6	Bin 2
7	Bin 2
8	Bin 2
9	Bin 3
10	Bin 3
11	Bin 3

(ii) Test Score
Minimum =40
Maximum =100
Range
=60
Bin Width

$$60/3 = 20$$

Intervals

Bin	**Interval**
Bin 1	40–60
Bin 2	60–80
Bin 3	80–100

Assignment

Score	**Bin**
40	Bin1
50	Bin1
55	Bin1
65	Bin2
75	Bin2
85	Bin3
90	Bin3
95	Bin3
98	Bin3

Score	Bin
100	Bin3

B. Equal-Frequency Binning
10 records
3 bins
Approximately
10/3 = 3–4 records each

Study Hours

Bin	Values
Bin1	2,3,4
Bin2	5,6,7
Bin3	8,9,10,11

Test Score

Bin	Values
Bin1	40,50,55
Bin2	65,75,85
Bin3	90,95,98,100

C. Concept Hierarchy
Study Hours

Hours	Category
2–4	Low
5–7	Medium
8–11	High

Test Score

Score	Category
40–59	Poor
60–79	Average
80–100	Excellent

Conclusion

- Equal-width binning divides values into equal-sized intervals.
- Equal-frequency binning places an equal number of records in each bin.
- Concept hierarchy converts numerical values into meaningful categories, making interpretation easier.

4) **Given Data**
200, 300, 400, 600, 1000

(a) Min-Max Normalization [0,1]
Formula

$$x' = \frac{x - min}{max - min}$$

Minimum =200
Maximum =1000
Range =800

Calculations
For 200

$$(200-200)/800 = 0$$

For 300

$$100/800 = 0.125$$

For 400

$$200/800 = 0.25$$

For 600

$$400/800 = 0.50$$

For 1000

$$800/800 = 1$$

Final Table

Original	Normalized
200	0.000
300	0.125
400	0.250
600	0.500
1000	1.000

(b) Z-Score Normalization
Formula

$$Z = \frac{x-\mu}{\sigma}$$

Mean

$$\mu = \frac{200+300+400+600+1000}{5} = 500$$

Standard Deviation
≈282.84

Calculations
200
=(200−500)/282.84
=-1.06
300
=-0.71
400
=-0.35
600
=0.35
1000
=1.77

Final Table

Original	Z-score
200	-1.06
300	-0.71
400	-0.35
600	0.35
1000	1.77

Comparison

Min-Max	Z-Score
Values between 0 and 1	Centered around 0
Simple scaling	Considers mean & standard deviation
Sensitive to outliers	Better for statistical analysis

5) **Dataset**
15, 30, 45, 60, 120, 300, 600, 900, 1800, 3600

A. Equal-Frequency Binning
10 values
3 bins
≈3–4 values per bin

Bins

Bin	Visit Duration (seconds)
Bin1	15,30,45
Bin2	60,120,300
Bin3	600,900,1800,3600

Analysis

- Bin 1 represents short visits.
- Bin 2 represents medium visits.
- Bin 3 represents long visits.
- Equal-frequency ensures each bin contains nearly the same number of customers despite the large range of durations.

B. Z-Score Normalization
Formula

$$Z = \frac{x - \mu}{\sigma}$$

Mean

$$\frac{7470}{10} = 747$$

Standard Deviation
≈1082

Z-Scores

Duration	Z-score
15	-0.68

Duration	Z-score
30	-0.66
45	-0.65
60	-0.64
120	-0.58
300	-0.41
600	-0.14
900	0.14
1800	0.97
3600	2.64

Discussion

- Most visit durations are clustered around the mean.
- The values **1800** and **3600 seconds** have high positive Z-scores, indicating unusually long visits.
- Z-score normalization preserves these extreme values while expressing them relative to the dataset.

C. Min-Max Normalization

Formula

$$x' = \frac{x - min}{max - min}$$

Minimum =15
Maximum =3600
Range =3585

Calculations

15
=0
30
=0.004
45
=0.008
60
=0.013
120
=0.029
300
=0.080
600
=0.163
900
=0.247
1800
=0.498
3600
=1.000

Final Table

Duration	Normalized Value
15	0.000
30	0.004
45	0.008

Duration	Normalized Value
60	0.013
120	0.029
300	0.080
600	0.163
900	0.247
1800	0.498
3600	1.000

Comparison of Z-Score and Min-Max

Feature	Min-Max Normalization	Z-Score Normalization
Output Range	0 to 1	Negative & Positive Values
Based On	Minimum & Maximum	Mean & Standard Deviation
Effect of Outliers	More sensitive	Less sensitive
Best Used For	Neural Networks, Distance-based algorithms	Regression, Statistical Analysis, Clustering